Day 3: Configuring Basic Network Hardware Theory (30 min): IP addressing basics and network configuration.Practical (30 min):Set up a router with default settings.

**1.Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' look for 'Default Gateway'. **

ANS:

Connect your laptop to your home Wi-Fi.

Open Command Prompt (CMD).

Type this command and press Enter:

CMD: ipconfig

Find Default Gateway under your Wi-Fi adapter. For example, 192.168.1.1.

Open Chrome and type that IP address in the address bar.

Press Enter. The router login page will open.



2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

ANS:

Steps:

1. Connect your laptop to Wi-Fi.
2. Open Command Prompt (CMD).
3. Type ipconfig and press Enter.
4. Find your Wi-Fi adapter and note the following details.

3.Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

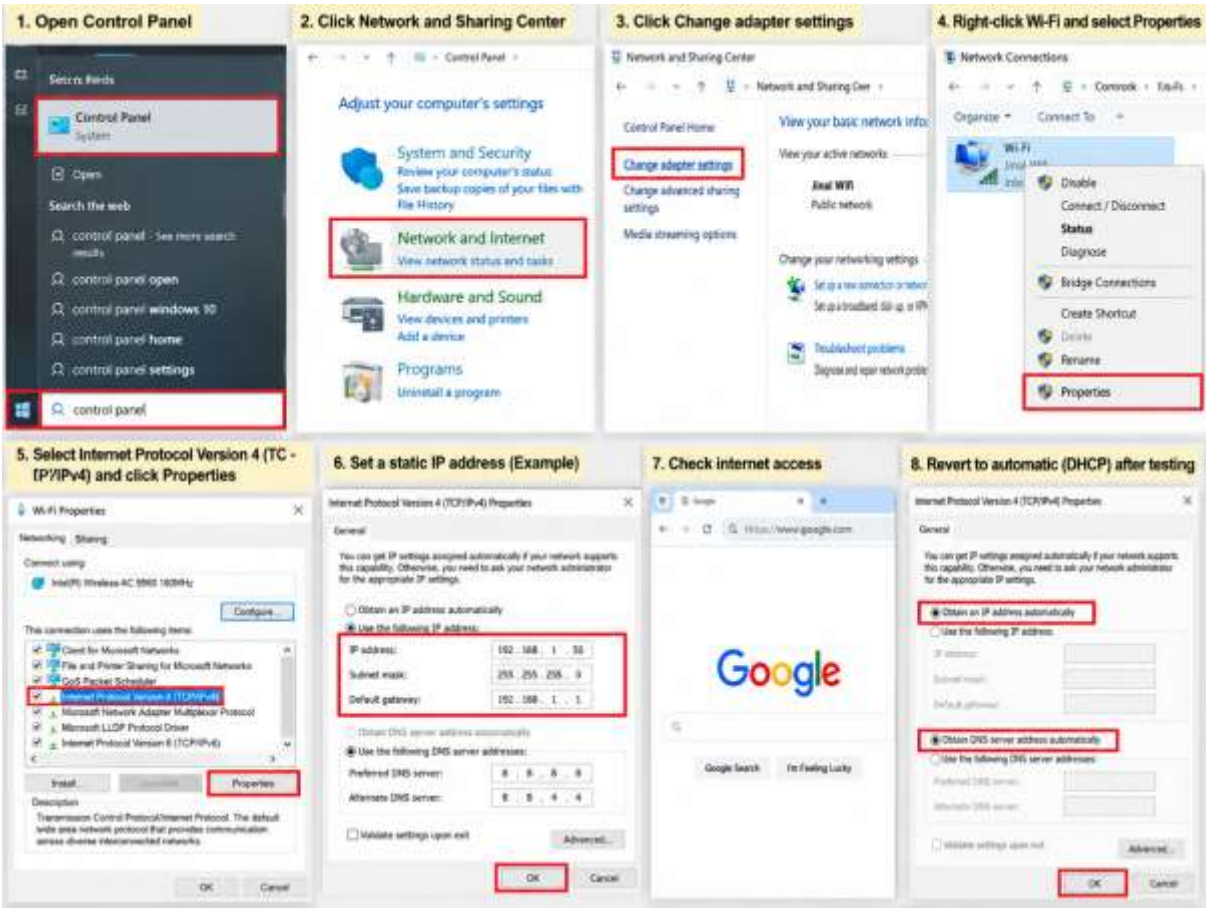
Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

ANS:

1. Open Control Panel.
2. Click Network and Sharing Center.
3. Click Change adapter settings.
4. Right-click Wi-Fi and click Properties.
5. Select Internet Protocol Version 4 (TCP/IPv4).
6. Select Use the following IP address.
7. Enter the IP address, subnet mask, and default gateway.
8. Click OK and check if the internet works.

Result: My laptop uses a static IP address, and I checked the internet connection.

Restore DHCP: Select Obtain an IP address automatically and click OK.



4.In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

ANS:

DHCP Range

Example: 192.168.1.100 to 192.168.1.200

Problems if two devices have the same IP address:

1. A network problem can happen.
2. The internet may stop working.
3. Devices may not connect properly.
4. Data may go to the wrong device.
5. We need to change one IP address.